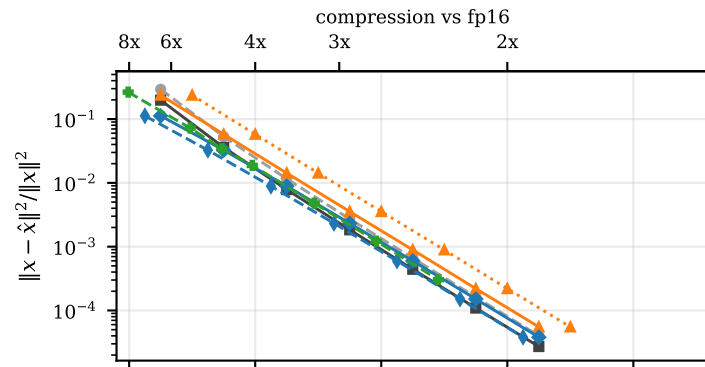
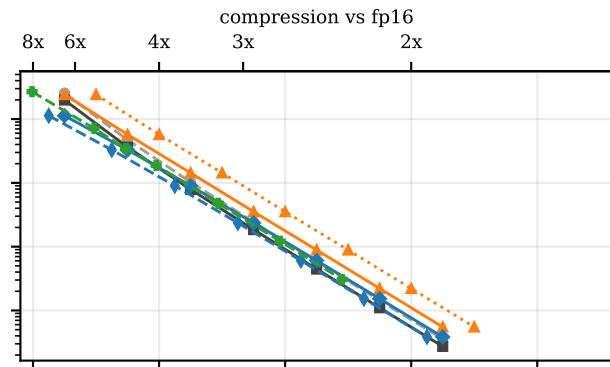


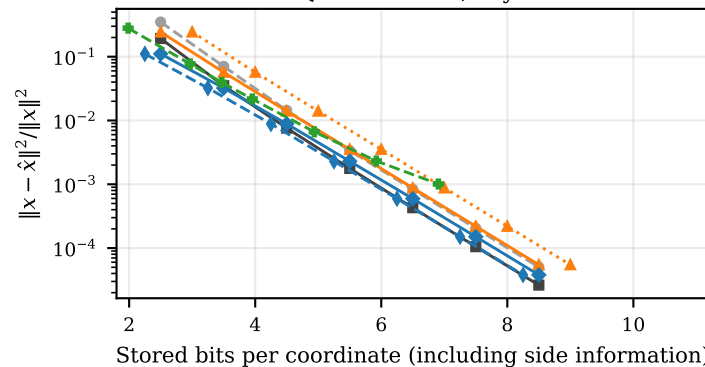
GPT-2, keys



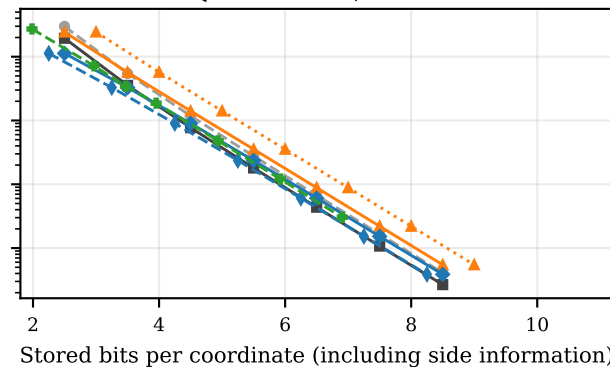
GPT-2, values



Qwen2.5-0.5B, keys



Qwen2.5-0.5B, values



—●— Round-to-nearest, no rotation

—▲— PITCH PolarQuant (single level)

—◆— TurboQuant-MSE, norm in fp16

—■— Rotate + round-to-nearest

—◆— PITCH TurboQuant-MSE

—+— Recursive PolarQuant (Python prototype)

—▲— PITCH PolarQuant, seed stored per vector